DELTA HIGHER INSTITUTE FOR MANAGEMENT
AND ACCOUNTING INFORMATION SYSTEMS

GENERATIVE ARTIFICIAL INTELLIGENCE AS AN ASSISTANT FOR DEVELOPERS

A Mini Research Report

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1. Introduction

Generative Artificial Intelligence (AI) is a technology that can create new content, including text, explanations, images, and computer code, in response to instructions from a user. Tools such as ChatGPT and Google Gemini can help beginners understand technical concepts and work through common web-development tasks. Rather than replacing developers, these tools can act as assistants for learning, planning, and problem-solving.

This report explains how generative AI can support web developers, especially beginners, through practical examples. It also discusses the main benefits, challenges, and responsible ways to use AI tools. The topic is relevant to Internet Applications and Electronic Platforms because digital tools and programming languages are important in creating and maintaining websites.

2. What Is Generative AI?

Generative AI refers to AI systems that produce new material in response to a prompt. A prompt is the question or instruction given by the user. For example, a learner might ask an AI assistant to explain HTML, compare CSS with JavaScript, or suggest a simple page layout. The system generates a response based on patterns learned during training, but its answer is not guaranteed to be correct.

Generative AI tools can produce explanations, summaries, sample code, and ideas. Their usefulness depends partly on how clearly the user describes the task. A specific prompt usually gives the system better guidance than a vague request such as “help me make a website.”

3. How AI Assists Web Developers

3.1 Explaining Code

Beginners may find unfamiliar code difficult to understand. An AI assistant can explain code line by line, define technical terms, and provide simpler examples. For instance, a student can paste a short HTML snippet and ask what each element means. Users should avoid sharing passwords, private information, or confidential code.

3.2 Generating Starter Code

AI can draft a basic example of HTML, CSS, or JavaScript from a clear description. A prompt might be: “Create a simple HTML page with a heading, a short paragraph, and a button.” The result can provide a starting point for practice, but the developer should read, test, and adapt the code rather than submit or publish it blindly.

3.3 Finding and Fixing Errors

When a page does not work as expected, an AI assistant may explain an error message or suggest possible fixes. For example, it may help identify a missing bracket in JavaScript or a spelling mistake in a CSS selector. Suggestions must be tested because the AI may misunderstand the problem or recommend a fix that causes another issue.

3.4 Planning and Improving Websites

AI can help organize a website into sections, suggest accessible labels, improve explanatory text, and produce a testing checklist. A developer might ask for ideas for a student-information page or an online store layout. Human judgment is still needed to choose suitable designs, protect user data, and ensure the site works for people with different needs.

4. Simple Practical Example

Imagine a beginner is creating a small student-club website. The student asks an AI assistant to suggest a page containing a title, an introduction, an activities list, and a contact button. The AI may provide starter HTML and CSS and explain how the parts work.

The student then opens the page in a browser, checks the layout on a phone, tests the button, corrects errors, and changes the content to match the club. If the button does not work, the student can ask for an explanation and then test any suggested change. This example shows AI being used as a learning partner while the student remains responsible for the final result.

5. Benefits of Generative AI for Developers

Faster learning: explanations and examples can help learners approach unfamiliar technical topics.

Quick drafts: starter code, outlines, and checklists can reduce repetitive work and help users begin a task.

Problem-solving support: AI may suggest ways to investigate errors and explain why a particular solution might work.

Idea generation: it can offer alternative layouts, wording, and approaches when a developer is planning a website.

6. Challenges and Responsible Use

AI-generated answers can contain incorrect information, insecure code, outdated advice, or references that do not exist. A response may sound confident even when it is wrong. For this reason, users should verify technical claims against reliable documentation, test code in a safe environment, and understand the result before using it.

Privacy is another important concern. Users should not enter passwords, personal records, private customer details, or confidential project information into an AI tool unless they have permission and understand the tool's data policies. Developers should also check accessibility, security, and compatibility instead of assuming that generated code is ready for publication.

In education, AI should support learning rather than replace independent thinking or honest academic work. Students should follow their instructor's rules, acknowledge AI assistance when required, and be able to explain the work they submit.

7. Conclusion

Generative AI tools such as ChatGPT and Gemini can help new web developers by explaining code, drafting examples, supporting debugging, and suggesting website improvements. Their value is greatest when users ask clear questions, review the answers, and test the results. AI should be treated as an assistant rather than an unquestionable authority. Combining AI support with reliable sources, practical testing, and human judgment can make learning web development more effective.

8. References

The following official or educational sources provide further information about AI tools and web development:

1. OpenAI. “ChatGPT.” Available at: https://chatgpt.com/ (Accessed 12 October 2026).

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